



How well does a current LLM understand Indirect Requests?

Introduction

- People often say things indirectly.
i.e. “It’s cold in here.”
- ❖ a literal statement?
 - ❖ an indirect request?
- Understanding the intended meaning can depend on what the speaker knows.

Speaker Unaware:

You and your friend are sweaty and thirsty after playing tennis. Back at your apartment, you privately learn that the water is shut off for 24 hours.
Friend: “I’m so thirsty...”

Speaker Aware:

You and your friend are sweaty and thirsty after playing tennis. Back at your apartment, your friend sees a notice saying the water is shut off for 24 hours.
Friend: “I’m so thirsty...”

Possible interpretations:

Request: “Could you get me a glass of water?”
Literal: “I’m really thirsty, but sadly your water isn’t working.”

- Trott & Bergen (2019):
On humans,
- ❖ Speaker Unaware: 76% request
 - ❖ Speaker Aware: 30% request

- Jones et al. (2024):
- ❖ GPT-3 did not show the same speaker-awareness effect.

Research Question

- RQ1:** Does GPT-5.6 Luna use speaker awareness when interpreting indirect requests?
- RQ2:** Does enabling reasoning change the speaker-awareness effect?

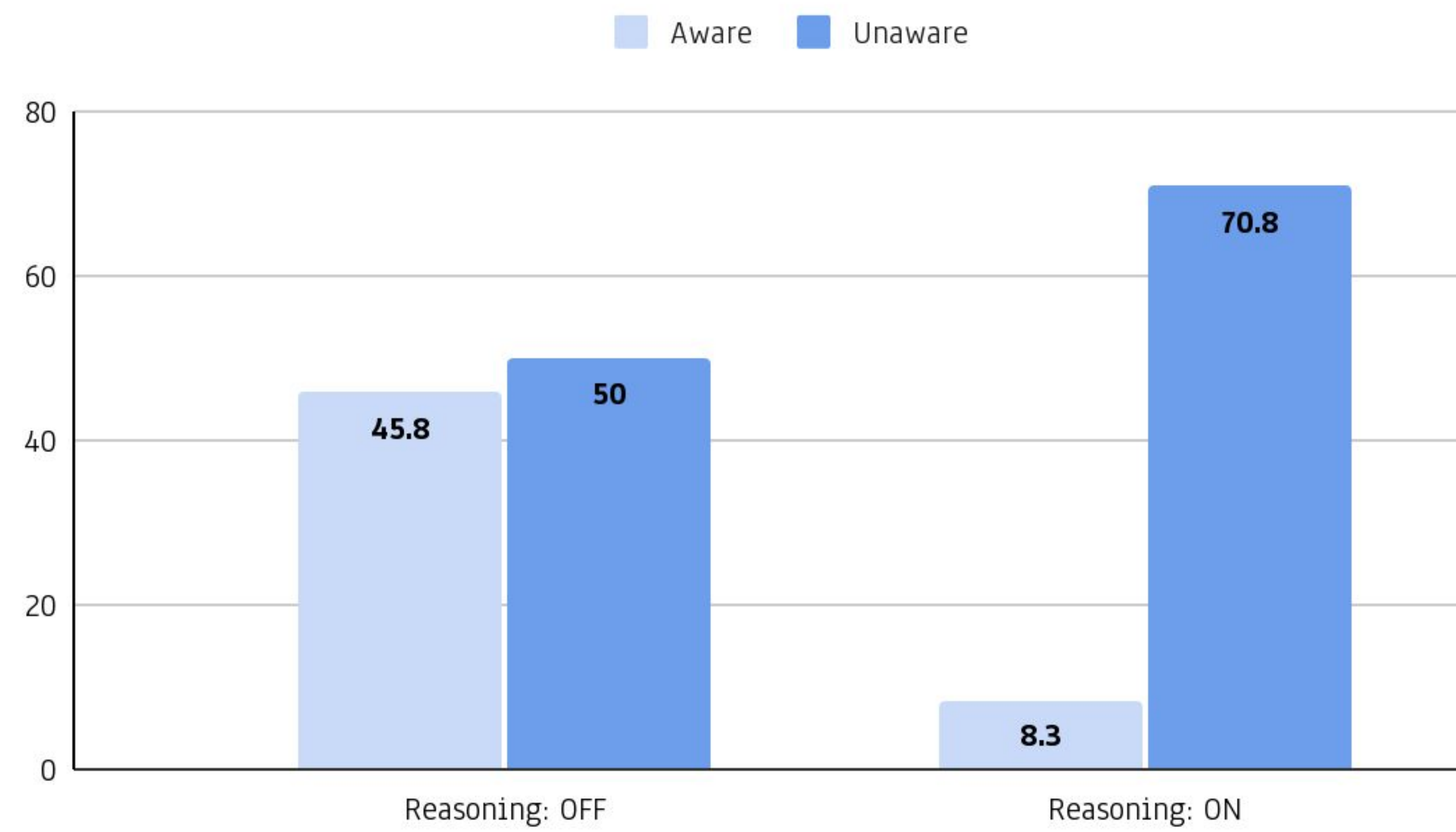
- H1:** More requests when Unaware
H2: Stronger effect with Reasoning ON

Method

- Materials
- ❖ 8 story pairs from Trott & Bergen (2019)
 - ❖ Model chose Request vs. Literal
 - ❖ Model: GPT-5.6 Luna
 - Reasoning OFF / ON (medium)
 - 3 runs per condition
 - 96 responses

Results

Request Interpretations (%)



- Fischer’s exact test:
OFF: +4.2 pp, $p = 1.000$
ON: +62.5 pp, $p < 0.001$
Reasoning x Awareness interaction: $p = 0.003$

Discussion

- ❖ OFF: little/no awareness effect
- ❖ ON: strong awareness effect, with a similar pattern to humans
- ❖ Shows Reasoning can strongly change how models use speaker knowledge

Limitations

- ❖ Only 1 LLM
- ❖ Only 8 story pairs
- ❖ Responses were not fully independent

References

Trott & Bergen (2019), Trott & Bergen (2020), Jones et al. (2024)